SOBEY SCHOOL OF BUSINESS
MASTER OF BUSINESS ANALYTICS
ACCOUNTING & FINANCE ANALYTICS (MBAN5570)
ASSIGNMENT - 3

Instructor: Mohammad M. Rahaman
Lecture Location: SB415
Meeting Time: M: 2.00PM-5.00PM

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DUE DATE: March 17, 2026 (Brightspace Dropbox)

Please read the following instructions carefully: This is your third assignment.Submit only one copy of your write-up. Clearly report your results and omit computational details unless explicitly requested. Provide brief explanations when necessary, and avoid embedding answers within large tables of numbers. Type all responses, both numerical and written, within a few pages. Submit only a .pdf file—do not include unedited output. Upload your submission to the designated Dropbox on Brightspace. All relevant data files for this assignment are available on the Brightspace Course Page. The assignment is graded out of 10, with all questions equally weighted.

I. PFIZER INC.: DEVELOPMENT OF A NEW DRAG

The pharmaceutical industry deals with a very high degree of uncertainty. Over 90% of all products under development fail to come to the market resulting in large loses. Products that do come to the market can earn multi-billion dollar profits annually for 10-15 years until the patent for the drug expires. In this exercise, you will analyze a new drug development decision by Pfizer Inc. and will utilize @RISK to model uncertainty associated with that decision to develop a new drag. The new drug being developed is called NIAGARA and before going to the marker NIAGARA must go through the following stages of development:

1. Initial Research & Development (R&D) Phase

2. Pre-Clinical Testing Phase
3. Clinical Trial - Phase I
4. Clinical Trial - Phase II

Only after all development stages succeed can the drug be sold to the general population. If the drug fails at any stage, then the development process is terminated all together. A success at any stage leads to pursue the next stage. You have to determine the risk-adjusted Net Present Value (NPV) from this new drug development project and investigate the key drivers of the new drug's profitability. You have to use the Triangular random variable from @RISK to model the following uncertain inputs at each stages of the development process: Cost, Probability of Success, and the Time Required to Complete the Stage. You will also model the PROFIT earned from the sale of the new drug as a Triangular random variable. The discount rate for this project is 15%. The input values are given below (note that the cost and profits are all-in cost/profit in millions of dollars):

	R&D Stage	**Pre-Clinical**	**Clinical-Pase I**	**Clinical-Phase 2**
Cost:				
Best	50.000	10.00	350.00	3500.00
Worst	120.00	30.00	600.00	6000.00
Most Likely	70.000	15.00	480.00	4200.00
Time:				
Best	3.00	0.50	3.00	3.00
Worst	7.00	3.00	6.00	6.00
Most Likely	4.00	1.00	4.00	4.00
Probability of Success:				
Best	0.42	0.60	0.60	0.96
Worst	0.20	0.30	0.40	0.70
Most Likely	0.35	0.50	0.50	0.90

Likely Profits Once Developed	
Best	60000
Worst	14000
Most Likely	18000

For example, Pre-Clinical testing will cost \$10 million in the best case, \$30 million in the worst case, and \$15 million in the most likely case. In the best case, the Pre-Clinical stage will last for 0.50 years; in the worst case, the Pre-Clinical stage will last for 3.0 years; and in the most-likely case, the Pre-Clinical stage will last for 1.0 years. Finally, in the best case, the Pre-Clinical stage will have a success probability of 60%; in the worst case, the Pre-Clinical stage will have a success probability of 30%; and in the most-likely case, the Pre-Clinical stage will have a success probability of 50%. We are assuming here that these random variables are independent of one another.

Based on these uncertain inputs, develop a financial model (typically known as an NPV model) for this new drug development. Report the mean, median, 5th, 25th, 50th, 75th, and 95th percentiles of your Net Present Value (NPV). Report the NPV empirical distribution and the Value-at-Risk charts after 5000 simulation. Which stage of the development process is the most critical input of your NPV model? Report the Tornedo graph of your sensitivity analysis.

II. TIME-SERIES DATA ANALYTICS

Use the file `Y2026_Assignment3_BitcoinPriceData.csv`. The dataset contains daily Bitcoin data including:

- Date
- Open
- High
- Low
- Close
- Adjusted Close
- Volume

Q1: Data Preparation and Exploratory Analysis

The sample covers multiple years of daily observations. Using the Bitcoin dataset:

(a) Import the data and arrange the observations in chronological order.

(b) Construct the following variables:

- Closing price P_t
- Log price $\ln(P_t)$
- Daily log return

$$r_t = 100 \times \ln\left(\frac{P_t}{P_{t-1}}\right)$$

- Daily log volume growth

$$g_t = 100 \times \ln\left(\frac{V_t}{V_{t-1}}\right)$$

(c) Plot the following series:

- Closing price
- Daily log return
- Trading volume

(d) Briefly comment on:

- Whether Bitcoin price appears stationary
- Whether returns appear more stationary than prices
- Whether volatility appears constant over time

Q2: Moving Average and Exponential Smoothing Forecasts

Using the **closing price** series:

(a) Construct forecasts using:

- 5-day moving average
- 20-day moving average

(b) Construct forecasts using **simple exponential smoothing** with $\alpha = 0.2$ and $\alpha = 0.8$. Assume the initial forecast equals the first observed closing price.

(c) Use the last 20% of observations as a test set. Compare the forecasting performance of:

- MA(5)
- MA(20)
- SES ($\alpha = 0.2$)
- SES ($\alpha = 0.8$)

using the following metrics:

- CFE
- MSE
- MAD
- MAPE

Which method performs best? Provide an economic explanation.

Q3: Autocorrelation and Stationarity

Using the **daily log return** series:

(a) Plot the ACF of returns up to lag 20.

(b) Plot the PACF of returns up to lag 20.

(c) Based on the ACF and PACF, discuss whether the return process resembles:

- White noise
- AR dynamics
- MA dynamics
- ARMA dynamics

Explain why financial prices are usually non-stationary while returns are often closer to stationary.

Q4: ARIMA Forecasting

(a) Estimate one of the following models:

- AR(1) model for returns
- ARIMA(1,1,0) for log prices
- ARIMA(0,1,1) for log prices

(b) Produce 1-step-ahead forecasts for the test sample.

(c) Compare forecast accuracy with the best smoothing model from Question 2.

(d) Interpret whether past price movements help predict future Bitcoin returns.

Q5: Volume and Volatility Clustering

(a) Define a volatility proxy using squared returns:

$$VP_t = r_t^2$$

(b) Plot r_t and r_t^2. Comment on volatility clustering.

(c) Plot the ACF of squared returns.

(d) Estimate the following regression:

$$r_t^2 = \gamma_0 + \gamma_1 r_{t-1}^2 + u_t$$

Interpret the coefficient γ_1.

(e) Estimate the following model:

$$r_t^2 = \delta_0 + \delta_1 g_t + \delta_2 |r_{t-1}| + \varepsilon_t$$

Interpret the coefficients. Based on your findings, discuss whether a GARCH model would be appropriate.